Term Test, Second Semester 2022

MATHS 350: Topology

(Time allowed: 50 MINUTES)

Each question is worth 10 marks.

1. Prove that a family $\mathcal{B}$ of subsets of a set X is a basis for a topology on X if and only if
   (a) for each $x \in X$ there exists $B \in \mathcal{B}$ such that $x \in B$, and
   (b) for each $B_1, B_2 \in \mathcal{B}$ and each $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

2. Let A be an indexing set and for each $\alpha \in A$ let X_α be a topological space. Prove that the Tychonoff topology is the weakest topology on $X = \prod_{\alpha \in A} X_\alpha$ for which each of the projections $\pi_\alpha : X \to X_\alpha$ is continuous.

3. Let X and Y be topological spaces. Suppose that Y is Hausdorff and $f : X \to Y$ is a continuous function.
   (a) Prove that the set of fixed points of f, the set $\{x \in X : f(x) = x\}$, is closed.
   (b) Prove that if f is one-to-one then X is Hausdorff.

4. Prove that every regular second countable space is normal.
